

Quantum Communications and Networks(QU.50013)

Exercise Set 1

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Fall 2026

Problem 1

The benefits of applying quantum technology come not from communication speeds, but rather the usefulness of sharing quantum states itself. **Quantum key distribution** (QKD) is one of them: to ensure computational security, we use the quantum mechanical principle that perfectly discriminating non-orthogonal states is impossible. Another is **quantum teleportation**, where unknown states can be sent far away, enabling distributed quantum computing systems. In both of these cases, qubits and ebits (entanglement) are the crucial quantum resources.

Problem 2

For a qubit in a pure state

$$|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle,$$

measurement probabilities can be determined using the Born rule:

$$P(0) = |\langle 0|\psi\rangle|^2 = \frac{3}{4}, \quad P(1) = |\langle 1|\psi\rangle|^2 = \frac{1}{4}.$$

Problem 3

An unknown quantum state cannot be simply read, copied, and retransmitted as classical information could. This is called the *no-cloning theorem*, and it stems from the fact that quantum measurement processes are destructive, i.e., measuring a system collapses its state into one of the measurement bases. Thus, quantum communication requires *quantum repeaters*, which help with retaining coherence in long-distance entanglements without amplifying signals.

Problem 4

No, local measurements on Alice's side cannot send any information to Bob. While entanglement allows for nonlocal *correlations* between separate measurements, it cannot create nonlocal *causations* between them. This can be readily seen in the reduced density matrix formulation, wherein Alice's local operations (unitary or measurement) does not affect Bob's reduced density matrix. Therefore, no matter what Alice does with her qubit, Bob's measurement outcomes are unaffected.